

FINAL MOCK ASSESSMENT

Time: 60 Minutes

Total Questions: 100

1. Which Python data type is mutable?
A. Integer
B. List
C. String
D. Tuple
Answer: B
2. Which SQL command is used to remove all records but keep the table structure?
A. DELETE
B. REMOVE
C. TRUNCATE
D. DROP
Answer: C
3. Which Pandas function displays the first five rows of a DataFrame?
A. display()
B. first()
C. head()
D. top()
Answer: C
4. What is the purpose of train-test split in Machine Learning?
A. Normalize data
B. Remove duplicates
C. Evaluate model performance
D. Increase dataset size
Answer: C
5. Which activation function is commonly used in hidden layers of deep neural networks?
A. Sigmoid
B. Linear
C. ReLU
D. Softmax
Answer: C
6. Which SQL clause is used to filter grouped records?
A. DISTINCT
B. HAVING
C. ORDER BY
D. WHERE
Answer: B
7. What is the output of `len("Artificial")`?
A. 11
B. 10
C. 9
D. 12
Answer: B

8. Which library is mainly used for numerical computation in Python?

- A. Seaborn
- B. Pandas
- C. NumPy
- D. Matplotlib

Answer: C

9. Which algorithm is used for classification?

- A. Logistic Regression
- B. Linear Regression
- C. PCA
- D. K-Means

Answer: A

10. What does OOP stand for?

- A. Open Object Processing
- B. Online Object Programming
- C. Object Oriented Process
- D. Object Oriented Programming

Answer: D

11. Which SQL function returns the total number of records?

- A. AVG()
- B. MAX()
- C. COUNT()
- D. SUM()

Answer: C

12. Which chart is best for comparing values across categories?

- A. Scatter Plot
- B. Pie Chart
- C. Bar Chart
- D. Line Chart

Answer: C

13. Which statement creates a new object in Python?

- A. new Student()
- B. Student()
- C. class Student()
- D. create Student()

Answer: B

14. Which evaluation metric is commonly used for classification?

- A. Accuracy
- B. RMSE
- C. MSE
- D. MAE

Answer: A

15. Which layer extracts image features in a CNN?

- A. Flatten Layer
- B. Pooling Layer
- C. Dense Layer
- D. Convolution Layer

Answer: D

16. Which Git command uploads local commits to GitHub?

- A. git commit

- B. git push
- C. git clone
- D. git add

Answer: B

17. Which function removes duplicate rows in Pandas?

- A. unique()
- B. remove_duplicates()
- C. delete_duplicates()
- D. drop_duplicates()

Answer: D

18. Which keyword is used for exception handling in Python?

- A. exception
- B. error
- C. try
- D. catch

Answer: C

19. Which SQL JOIN returns only matching records from both tables?

- A. FULL JOIN
- B. LEFT JOIN
- C. INNER JOIN
- D. RIGHT JOIN

Answer: C

20. Which preprocessing step removes common words like "the" and "is"?

- A. POS Tagging
- B. Stopword Removal
- C. Lemmatization
- D. Stemming

Answer: B

21. Which algorithm is commonly used for clustering?

- A. Decision Tree
- B. K-Means
- C. Linear Regression
- D. Logistic Regression

Answer: B

22. What is the purpose of a constructor in Python?

- A. Compare objects
- B. Initialize objects
- C. Print objects
- D. Delete objects

Answer: B

23. Which TensorFlow API is commonly used to build deep learning models?

- A. Tkinter
- B. OpenCV
- C. Keras
- D. Flask

Answer: C

24. Which SQL command is used to sort records?

- A. GROUP BY
- B. SORT BY
- C. ARRANGE BY

D. ORDER BY

Answer: D

25. Which NLP technique identifies words such as person names, locations, and organizations?

A. Tokenization

B. Named Entity Recognition

C. Stemming

D. POS Tagging

Answer: B

26. Which Python function is used to take input from the user?

A. read()

B. print()

C. input()

D. scan()

Answer: C

27. Which SQL constraint ensures that duplicate values are not allowed?

A. CHECK

B. FOREIGN KEY

C. PRIMARY KEY

D. UNIQUE

Answer: D

28. Which Pandas function is used to read a CSV file?

A. pd.load_csv()

B. csv.read()

C. open_csv()

D. pd.read_csv()

Answer: D

29. Which machine learning algorithm is commonly used for predicting continuous values?

A. Decision Tree Classifier

B. Linear Regression

C. K-Means

D. Logistic Regression

Answer: B

30. Which activation function is generally used in the output layer of binary classification?

A. Softmax

B. ReLU

C. Sigmoid

D. Tanh

Answer: C

31. Which SQL command is used to modify existing records?

A. CHANGE

B. MODIFY

C. ALTER

D. UPDATE

Answer: D

32. Which Python keyword is used to define a function?

A. define

B. func

C. function

D. def

Answer: D

33. Which library is mainly used for plotting graphs in Python?

A. TensorFlow

B. Matplotlib

C. Pandas

D. NumPy

Answer: B

34. Which of the following is an unsupervised learning algorithm?

A. Logistic Regression

B. Random Forest

C. Decision Tree

D. K-Means

Answer: D

35. Which OOP concept allows one class to acquire properties of another class?

A. Polymorphism

B. Abstraction

C. Inheritance

D. Encapsulation

Answer: C

36. Which SQL clause is used to remove duplicate rows from the result?

A. FILTER

B. GROUP BY

C. DISTINCT

D. UNIQUE

Answer: C

37. Which chart is best for showing trends over time?

A. Histogram

B. Pie Chart

C. Line Chart

D. Bar Chart

Answer: C

38. Which layer converts 2D feature maps into a 1D vector in CNN?

A. Pooling

B. Dense

C. Flatten

D. Convolution

Answer: C

39. Which Git command downloads a repository from GitHub?

A. git init

B. git pull

C. git fetch

D. git clone

Answer: D

40. Which NLP technique breaks text into words or sentences?

A. Parsing

B. Lemmatization

C. Tokenization

D. Chunking

Answer: C

41. Which function is used to display information about a DataFrame?

A. summary()

B. shape()

C. info()

D. describe()

Answer: C

42. Which evaluation metric is suitable for regression models?

A. Recall

B. Mean Squared Error (MSE)

C. Precision

D. Accuracy

Answer: B

43. Which SQL function returns the largest value in a column?

A. UPPER()

B. HIGH()

C. MAX()

D. TOP()

Answer: C

44. What is the default return value of a Python function if no return statement is used?

A. Empty String

B. None

C. False

D. 0

Answer: B

45. Which deep learning framework was developed by Google?

A. MXNet

B. Caffe

C. PyTorch

D. TensorFlow

Answer: D

46. Which statement is used to skip the current iteration of a loop?

A. return

B. continue

C. pass

D. break

Answer: B

47. Which SQL command permanently removes a table and its structure?

A. DELETE

B. REMOVE

C. DROP

D. TRUNCATE

Answer: C

48. Which Pandas function is used to remove missing values?

A. clean()

B. fillna()

C. dropna()

D. remove_na()

Answer: C

49. Which model is generally less prone to overfitting due to averaging multiple decision trees?
- A. Logistic Regression
 - B. KNN
 - C. Random Forest
 - D. Decision Tree
- Answer: C
50. Which OpenCV function is commonly used to read an image?
- A. cv2.read()
 - B. cv2.open()
 - C. cv2.load()
 - D. cv2.imread()
- Answer: D
51. Which Python statement is used to create an anonymous function?
- A. lambda
 - B. def
 - C. func
 - D. anonymous
- Answer: A
52. Which SQL function returns the current date?
- A. DATE()
 - B. CURDATE()
 - C. TODAY()
 - D. NOWDATE()
- Answer: B
53. Which Pandas function is used to combine two DataFrames vertically?
- A. merge()
 - B. concat()
 - C. join()
 - D. append_rows()
- Answer: B
54. Which preprocessing technique scales data between 0 and 1?
- A. Standardization
 - B. One-Hot Encoding
 - C. Normalization
 - D. Label Encoding
- Answer: C
55. Which of the following is a classification algorithm?
- A. Linear Regression
 - B. Logistic Regression
 - C. PCA
 - D. K-Means
- Answer: B
56. Which activation function is used in the output layer of multi-class classification?
- A. Sigmoid
 - B. ReLU
 - C. Tanh
 - D. Softmax
- Answer: D

57. Which SQL operator is used to search for a pattern in text?
- A. MATCH
 - B. REGEX
 - C. LIKE
 - D. FIND
- Answer: C
58. What will `range(2, 10, 2)` return?
- A. 2,3,4,5,6,7,8,9
 - B. 2,4,6,8
 - C. 1,3,5,7,9
 - D. 2,4,6,8,10
- Answer: B
59. Which Pandas function is used to find missing values?
- A. `missing()`
 - B. `isna()`
 - C. `null()`
 - D. `checkna()`
- Answer: B
60. Which algorithm is commonly used for dimensionality reduction?
- A. Decision Tree
 - B. PCA
 - C. Random Forest
 - D. Logistic Regression
- Answer: B
61. Which layer reduces feature map size in CNN?
- A. Dense Layer
 - B. Convolution Layer
 - C. Flatten Layer
 - D. Pooling Layer
- Answer: D
62. Which OOP concept hides internal implementation from users?
- A. Inheritance
 - B. Abstraction
 - C. Polymorphism
 - D. Association
- Answer: B
63. Which SQL clause groups rows having similar values?
- A. HAVING
 - B. GROUP BY
 - C. ORDER BY
 - D. DISTINCT
- Answer: B
64. Which evaluation metric considers both precision and recall?
- A. Accuracy
 - B. Recall
 - C. F1-Score
 - D. MAE
- Answer: C
65. Which Git command records changes in the local repository?
- A. `git push`

- B. git commit
- C. git clone
- D. git merge

Answer: B

66. Which NLP technique converts words like "running" into "run"?

- A. Tokenization
- B. POS Tagging
- C. Stemming
- D. Chunking

Answer: C

67. Which Python keyword is used to inherit a class?

- A. extends
- B. super
- C. inherit
- D. class(ClassName)

Answer: D

68. Which SQL JOIN returns all records from the left table and matching records from the right table?

- A. INNER JOIN
- B. FULL JOIN
- C. LEFT JOIN
- D. RIGHT JOIN

Answer: C

69. Which visualization is best for showing the relationship between two numerical variables?

- A. Pie Chart
- B. Scatter Plot
- C. Bar Chart
- D. Histogram

Answer: B

70. Which function is used to calculate correlation in Pandas?

- A. relationship()
- B. covariance()
- C. corr()
- D. correlate()

Answer: C

71. Which Machine Learning algorithm is based on a tree structure?

- A. KNN
- B. Decision Tree
- C. Naive Bayes
- D. Linear Regression

Answer: B

72. What is the main purpose of Dropout in Deep Learning?

- A. Increase neurons
- B. Reduce training time
- C. Prevent overfitting
- D. Normalize data

Answer: C

73. Which OpenCV function converts an image to grayscale?

- A. cv2.gray()

- B. cv2.cvtColor()
- C. cv2.cvtColor()
- D. cv2.cvtColor()

Answer: C

74. Which SQL function calculates the average value?

- A. SUM()
- B. COUNT()
- C. AVG()
- D. MEAN()

Answer: C

75. Which Python collection does not allow duplicate values?

- A. List
- B. Dictionary
- C. Tuple
- D. Set

Answer: D

76. Which of the following Python statements correctly creates a dictionary comprehension?

- A. {x:x*x for x in range(5)}
- B. [x:x*x for x in range(5)]
- C. (x:x*x for x in range(5))
- D. {x*x for x in range(5)}

Answer: A

77. Which SQL query returns departments having more than 5 employees?

- A. SELECT Department FROM Employee WHERE COUNT(*)>5;
- B. SELECT Department FROM Employee GROUP BY Department HAVING COUNT(*)>5;
- C. SELECT Department HAVING COUNT(*)>5;
- D. SELECT Department GROUP BY COUNT(*)>5;

Answer: B

78. In Pandas, which function is commonly used to apply a custom function to every element of a Series?

- A. map()
- B. apply()
- C. transform()
- D. replace()

Answer: B

79. Which machine learning algorithm is most sensitive to feature scaling?

- A. Decision Tree
- B. Random Forest
- C. K-Nearest Neighbors (KNN)
- D. Naive Bayes

Answer: C

80. Which evaluation metric should be preferred when classes are highly imbalanced?

- A. Accuracy
- B. Mean Squared Error
- C. F1-Score
- D. R² Score

Answer: C

81. In Deep Learning, vanishing gradients mainly affect which activation function?

- A. ReLU
- B. Leaky ReLU
- C. Softmax
- D. Sigmoid

Answer: D

82. Which optimizer adapts the learning rate during training?

- A. SGD
- B. Adam
- C. Batch Gradient Descent
- D. Newton Method

Answer: B

83. Which SQL command creates a relationship between two tables?

- A. CHECK Constraint
- B. UNIQUE Constraint
- C. FOREIGN KEY Constraint
- D. DEFAULT Constraint

Answer: C

84. In CNN, increasing the stride generally results in:

- A. Larger output feature maps
- B. Smaller output feature maps
- C. More trainable parameters
- D. Higher image resolution

Answer: B

85. Which Pandas function can simultaneously group and aggregate data?

- A. merge()
- B. groupby()
- C. sort_values()
- D. concat()

Answer: B

86. What is the main purpose of Batch Normalization?

- A. Increase dataset size
- B. Reduce overfitting only
- C. Normalize intermediate activations for stable training
- D. Replace activation functions

Answer: C

87. Which NLP technique represents words as dense numerical vectors?

- A. Tokenization
- B. POS Tagging
- C. Word Embeddings
- D. Stemming

Answer: C

88. Which OpenCV function detects edges using the Canny algorithm?

- A. cv2.edge()
- B. cv2.Canny()
- C. cv2.detectEdges()
- D. cv2.findEdges()

Answer: B

89. In Git, what is the purpose of `git pull`?

- A. Upload local changes

- B. Download and merge remote changes
- C. Delete remote branch
- D. Undo previous commit

Answer: B

90. Which Python feature allows one function to accept any number of positional arguments?

- A. `**kwargs`
- B. `*args`
- C. `args[]`
- D. `tuple()`

Answer: B

91. Which SQL function replaces NULL values with another value in MySQL?

- A. `NVL()`
- B. `COALESCE()`
- C. `IFNULL()`
- D. `ISNULL()`

Answer: C

92. Which Machine Learning algorithm is an ensemble technique?

- A. Logistic Regression
- B. Random Forest
- C. K-Means
- D. PCA

Answer: B

93. Which layer in an LSTM decides which information should be discarded?

- A. Input Gate
- B. Forget Gate
- C. Output Gate
- D. Hidden Layer

Answer: B

94. Which activation function produces probabilities whose sum equals 1?

- A. ReLU
- B. Tanh
- C. Softmax
- D. Sigmoid

Answer: C

95. Which Pandas function reshapes data using rows and columns similar to Excel Pivot Table?

- A. `groupby()`
- B. `melt()`
- C. `pivot_table()`
- D. `stack()`

Answer: C

96. Which statement about Decision Trees is TRUE?

- A. They require feature scaling.
- B. They can handle both numerical and categorical data.
- C. They always outperform Random Forest.
- D. They cannot perform classification.

Answer: B

97. In TensorFlow/Keras, what does `model.evaluate()` return?

- A. Only predictions

- B. Only accuracy
- C. Loss and evaluation metrics
- D. Training history

Answer: C

98. Which NLP task identifies words such as "Microsoft", "India", and "Virat Kohli" in text?

- A. Sentiment Analysis
- B. POS Tagging
- C. Named Entity Recognition
- D. Topic Modeling

Answer: C

99. Which SQL query correctly finds the second highest salary?

- A. `SELECT MAX(Salary) FROM Employee;`
- B. `SELECT MIN(Salary) FROM Employee;`
- C. `SELECT MAX(Salary) FROM Employee WHERE Salary < (SELECT MAX(Salary) FROM Employee);`
- D. `SELECT Salary FROM Employee LIMIT 2;`

Answer: C

100. During model development, if the training accuracy is 99% and testing accuracy is 70%, the model is most likely suffering from:

- A. Underfitting
- B. Data Normalization
- C. Overfitting
- D. High Bias

Answer: C